C8 Periodicity – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Period 3 Trends and Structures **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Periodicity				Repeating trends in properties across periods
2	Block containing Na and Mg				s
3	Block containing Al to Ar				p
4	Trend in atomic radius across Period 3				Decreases
5	Why atomic radius decreases across Period 3				More protons, same shielding
6	General trend in first IE across Period 3				Increases
7	Trend in electronegativity across Period 3				Increases
8	Structure of Na, Mg and Al				Giant metallic
9	Structure of Si				Giant covalent (macromolecular)
10	Structure of Ar				Monatomic

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Melting Points **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Period 3 element with highest melting point				Si
2	Period 3 element with lowest melting point				Ar
3	Formula of a phosphorus molecule				P_4
4	Formula of a sulfur molecule				S_8
5	Structure of phosphorus, sulfur and chlorine				Simple molecular
6	Why Si has a very high melting point				Many strong covalent bonds to break
7	Why Al has a higher melting point than Na				More delocalised electrons, higher ion charge
8	Why S_8 melts higher than P_4				Larger molecule, stronger van der Waals
9	Order of melting point for S_8, P_4, Cl_2, Ar				$S_8 > P_4 > Cl_2 > Ar$
10	Why Ar has the lowest melting point				Single atoms, weakest van der Waals

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.